Theory of p-adic Closed Strings

Paul H. Frampton and Hitoshi Nishino
Department of Physics and Astronomy, University of North Carolina, Chapel Hill, North Carolina 27599-3255
(Received 24 February 1989)

We discuss the N-particle amplitude for p-adic closed bosonic strings; an effective field theory is derived and then it is shown how the space-time field equations follow from a generalized scaling property in a p-adic σ model. The closed superstring is discussed p-adically at the four-particle level, and finally a speculation is made about how nonperturbative supersymmetry breaking in the superstring might be approached using p-adics.

PACS numbers: 11.17.+y, 11.10.Lm

Strings are a major weapon in the artillery of the theoretical physicist and are applied to physical situations as varied as galaxy formation in the early Universe, large organic molecules, superconductivity, and fundamental physics at the Planck length. In addition to their diverse uses in physical problems, strings have instigated new areas of mathematical physics by linking together diverse topics such as algebraic geometry, analytic number theory, singularity (or catastrophe) theory, category theory, etc.

The axiomatic foundation of a fundamental string theory is not well established. One promising approach towards this goal involves the use of p-adic number fields. Although p-adic strings do not yet appear to have any direct physical interpretation, their mathematical simplicity strongly suggests that they can provide a powerful tool for understanding the dynamics of string theory itself.

The most detailed work in p-adic strings has been done for the simplest open string. One purpose of the present Letter is to fill in the corresponding details for *closed* strings. The results obtained already for open p-adic strings include the generalized N-particle tree amplitudes,[1-3] the effective field theory for the ground-state tachyon,[3,4] and rederivation of the space-time field equations from a renormalization-group analysis of a p-adic two-dimensional field theory.[5,6] In the case of the closed p-adic string only a partial treatment has appeared: the four-particle tree amplitude[1,7] and the kinetic term of the two-dimensional field theory.[8,9]

Completing the same steps for the p-adic closed string is more than a repetitive exercise, because it gives a new approach to the p-adic superstring in which full permutational invariance among the external momenta is maintained in the tree amplitude. We shall derive a four-particle p-adic closed-superstring amplitude in this Letter.

Tree amplitude.—For the four-particle tree amplitude with closed bosonic strings we begin with the Shapiro integral representation[10] of the Virasoro amplitude.[11] With z a complex variable the amplitude is

$$A_4=\int d^2z\,|z|^{2A}|1-z|^{2B}, \tag{1}$$

with $A=-\frac{1}{2}\alpha_s-1$, $B=-\frac{1}{2}\alpha_t-1$, $\alpha_s+\alpha_t+\alpha_u=-2$. To obtain a p-adic counterpart, the complex field is replaced by the quadratically extended p-adic field $\mathbf{K}(\sqrt{\tau})$ so that $z=x+\sqrt{\tau}y$, $\bar{z}=x-\sqrt{\tau}y$, and x,y are in the p-adic field Q_p. It is then easiest to evaluate the integral by introducing p-adic polar coordinates[12] with $z=\rho t$, $\bar{t}t=1$, $\bar{\rho}\rho=c=\bar{z}z$. We then have (all norms are now p-adic)

$$\int_{\mathbf{K}(\sqrt{\tau})}dz=\int dc\int_{t\bar{t}=1}dt/|t|\,, \tag{2}$$

and in the p-adic counterpart of Eq. (1) one finds the angular integral over t is trivial: $\int_{\bar{t}t=1}dt/|t|=1$, and by dividing the c integration into regions $c<1$, $c=1$, and $c>1$, it is straightforward to find

$$A_4^{(p)}=\sum_{x=s,t,u}\frac{(1-p^{-1})(1-p^{(1/2)\alpha_x})}{1-p^{(1/2)\alpha_x}}+(1-2p^{-1})$$
$$=\prod_x\Gamma_p(-\tfrac{1}{2}\alpha_x)\,, \tag{3}$$

by using the mass-shell condition $\alpha_s+\alpha_t+\alpha_u=-2$, as usual.

To extend these results to the general N-particle amplitude one begins with the Shapiro amplitude[10] for the general case, replaces the complex field $\mathbf{C}$ by $\mathbf{K}(\sqrt{\tau})$, and again uses the p-adic polar coordinates. All the angular integrals are still of the trivial form, $\int_{\bar{t}t=1}dt/|t|=1$. The radial integrals are closely similar to the real integrals in the corresponding open-string case. This leads to Feynman rules where the propagator in general ξ gauge[4] is given by

$$\Pi_\xi^{\text{closed}}(k)=(1-p^{-1})\left[\frac{1}{1-p^{(1/2)\alpha(k^2)}}-\xi\right], \tag{4}$$

and the vertices V^m are identical to those of Refs. 2 and 4. For example, if $\xi=1$, one finds

$$V^m=(-1)^{m+1}\prod_{n=2}^{m-2}(n-p^{-1})\,, \tag{5}$$

for $m\geq 4$ and $V^3=1$.

The above is for the quadratic extension $\mathbf{K}(\sqrt{\tau})$ with $\tau=p$. There are two alternative extensions for general p, namely, $\tau = v_p$ and $\tau = v_p p$, where v_p is the smallest positive integer not equal (mod$p$) to any of the perfect squares 1,4,9,16,.... For these two alternative quadratic extensions, one finds that p in all p-adic formulas is simply replaced by p^2.

Effective field theory.— Since the Feynman rules for the N-particle tree amplitude with external closed-string tachyons are closely parallel, at the p-adic level, to those for external open-string tachyons, it is possible to write an effective Lagrangian which reproduces at tree level the correct S-matrix elements. The principal change from the analysis for open p-adic strings[2-4] is that in the kinetic term the d'Alembertian $\Box$ is replaced by $\Box/2$. The space-time field equation for the closed p-adic string is hence

$$p^{\Box/4}(1+\phi) = (1+\phi)^{1/p}. \tag{6}$$

The effective potential $V(\phi)$ can be computed explicitly and nonperturbatively. There is an unstable vacuum at $\phi = 0$ reflecting the presence of a tachyon. There is a shifted vacuum at $\phi = -1$ with no single-particle states but with positive-energy solitons analogous to those found in Ref. 3 for the open string.

The p-adic approach offers the only technique known to us of studying the string nonperturbatively. By analyzing the scattering of solitons one can, in principle, find the p-adic scattering amplitude for the true vacuum. While there exist static one-soliton solutions of Eq. (6), two-soliton solutions are necessarily time dependent because of the intersoliton force. This entails a mixing of time and space coordinates in the solution of the nonlinear field equation (6). If such an analytic solution could be found, it is possible that, by an adelic formula, one could reconstruct the actual (non-p-adic) string amplitude for the true vacuum. If such a calculation can be successfully completed, it will be very important and vindicate our faith in the p-adic technique. Until now, the only (purely technical) obstacle is to find the appropriate multisoliton solution of Eq. (6).

As we shall indicate below, the p-adic approach to the superstring will reveal very important information on, e.g., nonperturbative supersymmetry breaking, even without analytic solution of a nonlinear equation.

We should note that, as before, replacement of $\tau = p$ by $\tau = v_p$ or $v_p p$ in $\mathbf{K}(\sqrt{\tau})$ leads to replacement of p by p^2 in the effective Lagrangian.

σ model and vanishing β functions.— In Refs. 5 and 6 it was demonstrated how to set up a discretized p-adic renormalization group for a p-adic σ model such that the vanishing β function generated the space-time field equation for the open p-adic string.

For the closed p-adic string, we must reconsider the σ-model action.[8,9] For the open string one used[5]

$$S_{\text{open}} = -\frac{1}{2}\frac{p(p-1)}{p+1}\frac{1}{\ln p}\int_{Q_p} dx\,dy \frac{X^\mu(x)X_\mu(y)}{|x-y|_p^2} - \int_{Q_p} dx\, g\Phi[X(x)]. \tag{7}$$

For the closed case, it is necessary to start with

$$S_{\text{closed}} = -\frac{p^3(p-1)}{p+1}\frac{1}{\ln p}\int_{\mathbf{K}(\sqrt{p})} dz\,dz' \frac{X^\mu(z)X_\mu(z')}{|z-z'|_p^4} - \int_{\mathbf{K}(\sqrt{p})} dz\, g\Phi[X(z)]. \tag{8}$$

In Eq. (8), there is a scale invariance of the first term under $z \to pz$, $X^\mu(z) \to X^\mu(z)$, while the background term violates this symmetry as in the usual string case. Expanding around a background field X_0^μ by $X^\mu(z) = X_0^\mu + \xi^\mu(z)$, the path integral may be taken over $\xi^\mu(z)$ by rewriting Eq. (8) as

$$S_\xi = -\tfrac{1}{2}\int_{\mathbf{K}(\sqrt{p})} dz\, X^\mu(z)\Delta X_\mu(z) - \int_{\mathbf{K}(\sqrt{p})} dz\, g\sum_{n=1}^{\infty}\frac{1}{n!}\xi^{\mu_1}\cdots\xi^{\mu_n}\partial_{\mu_1}\cdots\partial_{\mu_n}\Phi(X_0). \tag{9}$$

The kinetic term has been chosen such that the ξ propagator is $\langle\xi^\mu(z)\xi^\nu(z')\rangle = -g_{\mu\nu}\ln|z-z'|$. An ultraviolet cutoff may be introduced by writing this as

$$\langle\xi^\mu(z)\xi^\nu(z')\rangle = \begin{cases} -\tfrac{1}{2}g_{\mu\nu}\ln|z-z'|^2, & \text{for } |z-z'|^2 \ge p^{-K}, \\ -\tfrac{1}{2}g_{\mu\nu}\ln p^{-K}, & \text{for } |z-z'|^2 < p^{-K}, \end{cases} \tag{10}$$

for some larger integer K.

Using this regularized propagator we may then calculate the effective action $W[\Phi]$ by

$$\exp\left(-\int_{\mathbf{K}(\sqrt{p})} dz\, W[\Phi]\right) = \exp\left(\int_{\mathbf{K}(\sqrt{p})} dz\, g\Phi(X_0)\right)\int[\mathcal{D}\xi]\exp\left(\tfrac{1}{2}\int_{\mathbf{K}(\sqrt{p})} dz\,\xi^\mu\Delta\xi_\mu\right)$$
$$\times\exp\left[g\int_{\mathbf{K}(\sqrt{p})} dz\sum_{n=1}^{\infty}\frac{1}{n!}\xi^{\mu_1}\cdots\xi^{\mu_n}\partial_{\mu_1}\cdots\partial_{\mu_n}\Phi(X_0)\right], \tag{11}$$

and hence expand $W[\Phi]=\sum_{N=1}^{\infty}W^{(N)}[\Phi]$ with $W^{(N)}$ purely of order Φ^N. After appropriate changes of variables the sum over Feynman graphs for fixed N (and cutoff K) gives

$$W^{(1)}=-\tilde{g}p^{K(\Box/4+1)}\Phi(X_0)\,, \tag{12}$$

$$W^{(2)}=-\tfrac{1}{2}\tilde{g}^2p^{K(\Box/4+1)}\left(\frac{1-p^{-1}}{1-p^{1-(1/2)\partial_1\cdot\partial_2}}+p^{-1}\right)\Phi^2\,, \tag{13}$$

with $\tilde{g}\equiv p^{-K}g$. Here $\Phi^2\equiv\Phi(X_1)\Phi(X_2)$, ∂_i acts on X_i, and after differentiation one puts $X_1=X_2$ $(=X_0)$; also $\Box=(\partial_1+\partial_2)^2$. More general $W^{(N)}$ are similar to that of Eq. (2.18) in Ref. 6, except that for the closed string $\partial_i\cdot\partial_j\to\frac{1}{2}\partial_i\cdot\partial_j$, etc.

To define a p-adic closed-string β function we write $-\tilde{g}\Phi_R(X_0)=\sum_{N=1}^{\infty}W^{(N)}$, as the renormalized Φ_R (Ref. 6) and regard the previous Φ as *bare* Φ_B; whereupon

$$\beta(\Phi_B)=[\Phi_B^{(K+1)}-\Phi_B^{(K)}]\,|_{\Phi_R\,\text{fixed}}\,. \tag{14}$$

One can then find that setting $\beta(\Phi_B)=0$ reproduces the field equation (6).

Replacement of $\tau=p$ by $\tau=v_p$ or v_pp in $\mathbf{K}(\sqrt{\tau})$ would mean that the prefactor in Eq. (8) be replaced by $-(p^2-1)/(2\ln p)$ and that in Eq. (13) p is everywhere replaced by p^2.

p-adic superstring.— Because the p-adic amplitudes for open and closed bosonic strings are attractively simple, one might expect that p-adic amplitudes exist in a simple form also for the superstring which is, after all, the only string for which perturbative finiteness can occur. Approaches to the SO(32) open superstring need to confront the fact that permutational invariance is lost due to the Chan-Paton factors.[3] For the closed superstring, for example, the type-II superstring, the situation is much more favorable at least for the four-particle level. The four-particle amplitude for four external massless bosons (gravitons, antisymmetric tensors, and dilatons) is [13]

$$A=\frac{\kappa^2}{128}\frac{\Gamma(-s/8)\Gamma(-t/8)\Gamma(-u/8)}{\Gamma(1+s/8)\Gamma(1+t/8)\Gamma(1+u/8)}\times\zeta_1^{\mu_1\mu_2}\zeta_2^{\nu_1\nu_2}\zeta_3^{\rho_1\rho_2}\zeta_4^{\sigma_1\sigma_2}K_{\mu_1\nu_1\rho_1\sigma_1}K_{\mu_2\nu_2\rho_2\sigma_2}\,, \tag{15}$$

where $K_{\mu\nu\rho\sigma}$ is a totally symmetric tensor in the external momenta k_i;

$$K_{\mu\nu\rho\sigma}\equiv k_{1\alpha}k_{2\beta}k_{3\gamma}k_{4\delta}\operatorname{tr}(R_0^{\mu\alpha}R_0^{\nu\beta}R_0^{\rho\gamma}R_0^{\sigma\delta})\,, \tag{16}$$

with $R_0^{\mu\nu}$ an antisymmetric second-rank tensor in the notation of Ref. 13.

For the special case of four external massless scalars (dilatons) one finds that Eq. (15) becomes

$$A=C(s^2+t^2+u^2)^2\prod_{\substack{x,y,z=s,t,u\\ \text{cyclic}}}\frac{\Gamma(1-\frac{1}{2}\alpha_x)}{\Gamma(3-\frac{1}{2}\alpha_y-\frac{1}{2}\alpha_z)}\,, \tag{17}$$

where C is a constant and $\alpha_x=2+\frac{1}{4}x$.

Equation (17) is not quite coincident with the basic Virasoro form which is

$$\int d^2z\,|z|^{2A}|1-z|^{2B}=\frac{\pi\Gamma(A+1)\Gamma(B+1)\Gamma(-1-A-B)}{\Gamma(A+B+2)\Gamma(-A)\Gamma(-B)}\,, \tag{18}$$

as we see by noting that in Eq. (17), $\alpha_s+\alpha_t+\alpha_u=6$. Nevertheless, it is desirable to rewrite Eq. (17) such that it has the form of Eq. (18) whereupon we may use the parallel p-adic forms used above for the bosonic closed string. Recalling that from above, we have

$$\int_{\mathbf{K}(\sqrt{p})}dz\,|z|^{2A}|1-z|^{2B}=\Gamma_p(A+1)\Gamma_p(B+1)\Gamma_p(-1-A-B)\,, \tag{19}$$

so that rewriting Eq. (17) in the form of Eq. (18) [and hence allowing the p-adic form Eq. (19)] will give the desired result.

To accomplish this goal, we may rewrite Eq. (17) as

$$A=-\tfrac{64}{3}C(s^2+t^2+u^2)\left(\frac{\Gamma(2-\frac{1}{2}\alpha_s)\Gamma(1-\frac{1}{2}\alpha_t)\Gamma(1-\frac{1}{2}\alpha_u)}{\Gamma(3-\frac{1}{2}\alpha_s-\frac{1}{2}\alpha_t)\Gamma(2-\frac{1}{2}\alpha_t-\frac{1}{2}\alpha_u)\Gamma(3-\frac{1}{2}\alpha_u-\frac{1}{2}\alpha_s)}+2\text{ perms}\right). \tag{20}$$

This is now in the required form and allows us to use the usual transition from $\mathbf{C}$ to $\mathbf{K}(\sqrt{p})$ to arrive at

$$A^{(p)}=-\tfrac{64}{3}C(s^2+t^2+u^2)\left[\sum_x(1-p^{-1})\left(\frac{2p^{(\alpha_x-2)/2}}{1-p^{(\alpha_x-2)/2}}+\frac{p^{(\alpha_x-4)/2}}{1-p^{(\alpha_x-4)/2}}\right)+3(1-2p^{-1})\right]. \tag{21}$$

The kinetic factor $s^2+t^2+u^2$ gives the spin structure for the graviton and scalar coupling to the external scalars precisely as in the superstring. The intermediate state has a coupling which decomposes into Legendre polynomials of the scattering angle z_s as

$$s^2+t^2+u^2\underset{s\to 0}{\approx}\tfrac{1}{3}s[P_2(z_s)+5P_0(z_s)]\,. \tag{22}$$

The overall kinematic factor is similar in spirit to the p-adic models discussed in Ref. 14. In the present case the coupling vanishes at precisely $s=0$, as is expected since the graviton and scalar couple to the stress-energy tensor.

Note the following.

(i) The (small) price for adding three terms in Eq. (21) is that spin-2 and -0 massive poles at $\alpha_x=4$ appear in the p-adic amplitude.

(ii) To establish an effective field theory for the p-adic superstring case, one needs only the superstring tree amplitudes for any number of external massless scalars. If the p-adic counterpart retains the simple pattern of the bosonic case, as seems to us likely, then this will provide a closed-form effective potential for the dilaton field ϕ_d. Just as one can see explicitly how the tachyon field ϕ is displaced to the shifted value $\phi=-1$ in the p-adic bosonic string theory, so one will then be able to ascertain whether $\phi_d=0$ is stable in superstring theory or whether the dilaton ϕ_d acquires mass. If it does acquire mass then supersymmetry has been broken, because the graviton is massless. We should emphasize that this exercise, which involves infinite summation over the number of external legs is nonperturbative and can provide the first access, known to us, to the question of dynamical supersymmetry breaking in superstrings.

(iii) While completing this manuscript, we received an interesting paper,[15] which derives alternative four-particle p-adic *open* bosonic string and superstring amplitudes by requiring simple adelic-type formulas. One significant difference between the results of this adelic requirement and our present results can be seen in the four-particle closed-superstring amplitude. In terms of the Gel'fand Γ function $\Gamma_q^{(p)}$,[12] we can derive for this an *alternative* amplitude

$$K\prod_{x=s,t,u}\Gamma_1^{(p)}(-x/8)\Gamma_{-1}^{(p)}(-x/8)\,, \tag{23}$$

where K is the kinematic factor of Eqs. (14) and (15). This has a simple adelic relationship with Eq. (15). Amplitude (23) has, however, multiple poles on the mass shell, no additive form analogous to Eq. (21), and hence cannot either be interpreted as arising from an effective field theory, or be used for the type of nonperturbative analysis proposed in note (ii) above. Only time will tell whether the effective field theory or the adelic formula is more important for understanding strings from the number-theoretic approach.

This work is supported in part by the U.S. Department of Energy under Grant No. DE-FG05-85ER-40219.

[1]P. G. O. Freund and E. Witten, Phys. Lett. B **190**, 191 (1987); see also I. Volovich, Classical Quantum Gravity **4**, L83 (1987); B. Grossman, Phys. Lett. B **191**, 101 (1987).

[2]P. H. Frampton and Y. Okada, Phys. Rev. Lett. **60**, 484 (1987).

[3]L. Brekke, P. G. O. Freund, M. Olson, and E. Witten, Nucl. Phys. **B302**, 305 (1988).

[4]P. H. Frampton and Y. Okada, Phys. Rev. D **37**, 3077 (1988).

[5]H. Nishino and Y. Okada, University of North Carolina, Chapel Hill, Report No. IFP-332-UNC, 1988 (unpublished); Phys. Lett. B (to be published); see also Y. Okada and M. R. Ubriaco, Phys. Rev. Lett. **61**, 1910 (1988).

[6]H. Nishino, Y. Okada, and M. R. Ubriaco, University of North Carolina, Chapel Hill, Report No. IFP-34-UNC (unpublished).

[7]P. G. O. Freund and M. Olson, Phys. Lett. B **199**, 186 (1987).

[8]R. B. Zhang, Phys. Lett. B **209**, 229 (1988).

[9]B. L. Spokoiny, Phys. Lett. B **208**, 401 (1988).

[10]J. A. Shapiro, Phys. Lett. **33B**, 361 (1970).

[11]M. A. Virasoro, Phys. Rev. **177**, 2309 (1969).

[12]I. M. Gel'fand, M. I. Graev, and I. I. Pyatetskii-Shapiro, *Representation Theory and Automorphic Functions* (Saunders, Philadelphia, 1968), p. 133.

[13]J. H. Schwarz, Phys. Rep. **89**, 223 (1982).

[14]P. H. Frampton, Y. Okada, and M. R. Ubriaco, Phys. Rev. D **39**, 1152 (1989).

[15]Ph. Ruelle, E. Thiran, D. Verstegen, and J. Weyers, Louvain Report No. UCL-IPT-89-01, 1989 (to be published).